

Date 10 Mar 2025

Telnet

SSH

Cloud Computing.

on demand delivery of ~~services~~ IT resources
over the internet with pay as you go pricing.

- Google
- Microsoft Azure
- IBM
- (AWS) Amazon

Services

SAAS

- drive

PAAS

- Environment

IAAS.

- Databases

Data Center.

Physical room or facility or building that
houses IT infrastructure building / running / delivery
application services

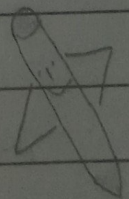
Virtualization.

Is a technology that uses software to
create virtual version of software & hardware.

* Question of Subnetting *

+ Cloud Models.

- public
- private
- Hybrid



Date 17/03/2025

→ (in-house data center)
On-Premises → physical
+ Service diagram (comparison)

Dedicated Servers

App. having its own dedicated server

- Accountability Authorization Authentication (AAA Server).
Manage credentials.

Server Virtualization.

Network interface card

Advantages.

NICs

→ b/w OS & firmware.

- Hypervisor is a s/w that creates virtual machine installed on hardware (eg. VMware, MS Hyper-V).
virtualized, enables multiple applications to run on single server.

- Multiple app. of single OS. → Container.

↓ container engine (Docker)

- Machine virtualization. → Multiple servers on one server.
- Container. → Multiple apps on one OS.

Date _____

Lecture - 1.

Cloud Computing.

On-demand availability of computer system resources,

especially computer power and storage

Storing & accessing data & programs over the internet.

Cloud Service Providers.

- Amazon web services.
- Microsoft Azure
- Google Cloud Platform
- IBM Cloud.

Cloud Models.

openstack.

• Private cloud

Services used by single org.

Complete control, meet specific business needs

google cloud, aws

• Public cloud.

Owned & operated by

3rd party cloud service

provider via internet



aws.

• Hybrid cloud.

Some servers on premises & extend

some to cloud.

Cloud Services.

- SaaS
- PaaS
- IaaS

Software distributed Model.

usually for app. development,

computing resources over internet

e.g. Digital Ocean, Google Compute engine,

Dropbox, Canva, office 365

tools over internet.

Amazon EC2, rackpace

Heroku, Google

App Engine.

• on - Premises.

Application

Data.

Runtime

Middleware

O/S

Virtualization

Networking.

Storage

Servers

✓ On premises ✓ IaaS ✓ PaaS ✓ SaaS

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SHOKIA
PRODUCTS

Signature _____

Date 7/4/25

Lecture - 3.

- Servers are computers that provides "services" to "clients".
 - Designed to service large no. of requests.
 - Org. typically require many physical servers to provide various services. (web, email, db, etc)

What is Microsoft Azure
definition [Imp.]

• Azure Resources & Resource Groups



Grouping of resources

Lecture 11

14-04-2025

• Azure subscription

(Accountability, Authorization, Authentication)



Dev → Test → Production.

Subscriptions

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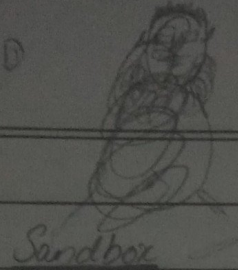
15-04-25

Lecture - 12.

SSH command.

- Big Data (large pools of data: that can be captured, communicated, aggregated, stored, & analyzed.)
- What we can do with data:
 - Store • Share • Access • Process • Encrypt





Date 24-04-2021

Lecture - 16

Sandbox

- definition → Important ← Availability Zones
with the help of diagram

leave slide - 14

Important

- Benefits of virtualization
- Types

- Network Virtualization
- Storage Virtualization
- RAID

Implementation
of virtualization

PDF

leave 19, 28, 29

Architecture of Cloud Computing
PDF 80.

- cloud [AWS, Oracle, Azure]
 - PDF 83, 84
- ↓ till mid.